Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/18/2026
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	VNSCOR: Responses of the Nervous System to Chronic, Low Dose Charged Particle Irradiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response (3) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grnelson@llu.edu	**Fax:**	FY 909-558-4035
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8364
Organization Name:	Loma Linda University		
PI Address 1:	Basic Sciences, Div. Radiation Research 11175 Campus Street		
PI Address 2:	Chan Shun Pavilion, Room A-1024		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	04/15/2018	**End Date:**	12/31/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per JSC Grants Office (Ed., 1/21/25) NOTE: End date changed to 12/31/2024 per JSC Grants Office (Ed., 4/26/23) NOTE: End date changed to 08/31/2025 per L. Juliette/JSC (Ed., 5/7/22)		
Key Personnel Changes/Previous PI:	Subcontract by Susanna Rosi, Ph.D. at the University of California San Francisco has been completed. Per the PI, the setup of behavioral experiments with Dr. Richard Hartman was completed by end of Year 2; after Year 2, Dr. Hartman left the project (Ed., 5/21/23).		
COI Name (Institution):	Mao, Xiao Wen M.D. (Loma Linda University) Rosi, Susanna Ph.D. (University of California San Francisco)		
Grant/Contract No.:	80NSSC18K0785		
Performance Goal No.:			

Performance Goal Text:

Task Description:

This project is a combined experimental campaign with "Mechanisms of Radiation-Induced Neurobehavioral Deficits" (PI: C. Davis) to quantify responses for an interrelated set of central nervous system (CNS) outcome measures in mice to acute and protracted exposures to protons, simulated galactic cosmic rays and gamma rays. The primary products are scaling factors DREF (dose rate effectiveness factor) and RBE (relative biological effectiveness) derived from animal studies which are used in radiation protection frameworks to guide estimates of the potential effects of space-like charged particle radiation delivered at low dose rates, from acute exposures of humans to gamma rays and x-rays, which are the main sources of evidence for radiation effects on humans.
An initial definition phase resulted in modifications to the original experimental plan to take advantage of new irradiation capabilities and to coordinate approaches with the Davis project by incorporating both projects into a virtual NSCOR or VNSCOR program project. Ongoing project revisions in collaboration with NASA were necessary to manage Covid crisis and physics facility scheduling issues. The start date for the project is April 15, 2018, and the period of performance was ultimately extended to December 31, 2025 to enable behavioral, transcriptomic and histological data analysis for animals irradiated during the experimental campaigns at Brookhaven National Laboratory. Statements of work were also adjusted in coordination with NASA science teams.

Evidence has accumulated from animal studies that the central nervous system (CNS) undergoes deleterious changes after exposure to charged particle radiation such as protons and high atomic number nuclei that are found in space as galactic cosmic rays and solar particle events. Observed changes include inflammation, oxidative stress, loss of neuron (dendrite) branches and connections (synapses), altered signaling molecules, altered electrical properties, loss of blood vessels, and impaired behavioral performance. If humans respond to charged particles in the same way as animals, then it is possible that deleterious changes may be sufficient to cause cognitive and other behavioral impairments that could compromise spaceflight missions and astronaut health. The current evidence for humans is based primarily on short exposures to single radiation types. However, space radiation is a complex mixture of these particles and exposures accumulate gradually over the course of missions. It is well established in radiation biology that reduction of the dose rate can have a profound effect on the outcome, especially for genotoxicity-related measures. Therefore, to better simulate the space environment, we exposed adult mice to either protons from the Loma Linda University synchrotron facility or mixtures of charged particles using the NASA/ Brookhaven National Laboratory (BNL)-developed 33-ion galactic cosmic ray simulation protocol (GCRsim) at the NASA Space Radiation Laboratory NSRL). The radiation was delivered in either in a single exposure or over 4 weeks in 24 short exposures (fractions) compatible with particle accelerator operations. These results were be compared to establish the Dose Rate Effectiveness Factors (DREFs) which are needed for risk estimation for astronaut health. We predicted that the high numbers (fluence) of protons will result in multiple traversals of cells within short times that may elicit interacting biological responses, whereas the lower fluence of higher charged ions would result in rare independent events. DREFs > 1 were predicted for protons and DREFs ~1 were predicted for high Z particles. We also compared equal doses of "chronic" or fractionated exposures of charged particle mixtures to gamma rays to determine whether they result in equal effect sizes or whether the charged particle mixtures were more effective. The relative biological effectiveness factors (RBEs) were derived from this comparison. These RBEs are utilized in predicting densely ionizing radiation effects in humans for whom only gamma ray and X-ray data are available with the assumption that the ratios obtained in animal models are realistic surrogates for humans. For this project, mice were irradiated with a broad energy spectrum of protons in acute and protracted (12 fractions over 4 weeks) exposures at a dose of 0.5 Gy and sham controls; acute and protracted (24 fractions over 4 weeks) exposures to 0.5 Gy of charged particles (33 ion GCR simulation or GCRsim); and protracted (24 fractions over 4 weeks) exposures to 2.0 Gy of 137-Cs gamma rays. An additional set of acute exposures to GCRsim was conducted to determine the form of the dose-response with acute exposures to 0, 0.15, 0.25, 0.5 and 0.75 Gy and an acute 137-Cs gamma ray exposure. All work used wild type mice and was performed under Institutional Animal Care and Use Committee (IACUC) approved protocols in AAALAC-certified facilities at Loma Linda University (LLU), the University of California (UCSF), and Brookhaven National Laboratory (BNL). For all aims the species is Mus musculus, strain C57Bl/6J. Ages were 5 - 6 months at acquisition and the beginning of irradiation procedures. We tested both male and female animals as their responses are not identical, and the astronaut population is of mixed gender. Scheduled sacrifices and tissue harvests followed behavioral testing. For each of the exposure regimens we conduct a battery of behavior tests, explored task-driven neuronal pathway activation patterns using c-fos imaging, quantified changes in selected gene expression patterns, and quantified selected neuroinflammation biomarkers using histochemical and microscopy methods. This allowed us to identify the underlying physiological changes that are most sensitive to dose rate and radiation quality and how they combined to produce behaviors that are adaptive or maladaptive. The Covid-19 pandemic disrupted the BNL experimental campaign resulting in >1-year delays in implementing GCR exposures. Therefore, during this period, exploratory studies of chronic mild stress were initiated to model multiple spaceflight stressors (e.g., altered gravity, isolation and confinement, sleep disruption). Findings from this pilot can enable future experiments using radiation exposures in combination with chronic stress to test interactions of multiple stressors with radiation. When BNL operations were restored we were able to expose male and female animals to GCRsim in the 2021, 2022 and 2023 campaigns.

GCRsim behavioral data on males and females has now been analyzed and DREF and RBE values calculated. Quantitation of c-fos expression after behavioral stimulation of GCRsim irradiated animals was used to map patterns of correlated expression between brain regions as measures of network activity and how the network activity associated with a fear memory behavioral task was modulated by prior radiation exposure. Assessment of glial cell activation using IBA-1 and GFAP markers was used to inform how neuroinflammation correlates with behavioral outcomes. Transcriptomics analysis was performed on GCRsim-irradiated animals to complement prior measurements with proton-irradiated animals. Results indicate that expression patterns (for a series of 760 genes associated with neuropathology) depend on brain region, sex, dose rate and type of radiation. A variety of statistical methods were used to transform data into a standard metric and to estimate effects sizes. These effect sizes were then used for estimation of DREF and RBE values from multiple independent measurements. This also enabled integrated behavioral assessments for domains such as exploration & locomotion, memory, anxiety-like behavior and sensorimotor performance using effect sizes and z-scores to express outcome measures in standard metrics. Together the data generated by the project will enhance NASA's ability to translate animal assessments of CNS (central nervous system) structure and function to humans, and to update risk estimates based on single radiation species, high dose rate irradiation protocols, to higher fidelity space-like exposures of charged particle mixtures delivered at dose rates approaching those observed in space.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary research impact to NASA is in estimation of potential adverse cognitive and behavioral effects of exposures to space radiation on long (e.g., 3-year Mars missions) deep space missions where exposures are of a chronic nature and are comprised of complex mixtures of charged particles dominated by protons. Such adverse effects could affect in-mission performance as well as post mission health of crew members. The experimental plan used radiation fields and exposure conditions scaled to the life span of the experimental animal model. On Earth, the principal benefits are for estimation of health risks to humans from charged particles in the environment (e.g., Radon alpha particles) and potential side effects of particle-based radiotherapy (e.g., proton and carbon beams) which requires pre-clinical animal studies. The task results provide insight into adverse effects on normal brain tissue from proton exposures similar to those expected for normal tissues outside tumor treatment volumes in head & neck and brain tumor treatment scenarios. Low dose rate exposures to protons as well as 33-ion GCRsim radiation fields with substantial proton and helium components will also inform risk estimates for the general population due to Radon exposures, internal emitter contaminations, and for first responders to radiological accidents (e.g., Fukushima, Chernobyl).
Task Progress:	We were able to meet the major goals of this project, but there were impacts from revisions to the original design. The use of protons early in the project enabled us to vet behavioral methods and select a suite of tests with verified sensitivity to radiation exposure. This was essential for DREF estimation and guided us to the use of effect sizes for the intercomparison of behavioral outcome measures. The replacement of electrophysiology by c-fos mapping was a positive revision and allowed for a more global assessment of network structure than a local neuron connectivity assessment could have provided, but missed the chance to directly quantify parameters associated with synaptic plasticity. Use of transcriptomics instead of a suite of Western blot and ELISA assays allowed us to broaden the survey of biomarkers that characterize the state of the brain tissue, and was a big labor and cost savings. The chronic mild stress pilot was successful and provides NASA with an option for simulating non-radiation flight stressors, and we were able to look at a blood-brain barrier surrogate by using the retinas of eyes from proton-irradiated mice. Use of a more balanced proportion of males and females than originally planned proved very informative and highlighted significant sex differences. A number of new questions arose from the project's findings, and we were able to test our initial hypotheses. We hypothesized that major drivers of the radiation response would be related to neuroinflammation and microglial activation, accompanied by damage to membrane structures and synaptic remodeling, but with only a small contribution from cell death. From theoretical concerns, we expected a DREF value > 1 for protons but ≈ 1 for GCRsim. We were prepared to see very large RBE values from the limited literature using different methods than our own. Mirroring traditional radiobiology thinking for consequences of dose rate reduction, we expected fractionated exposures to result in reductions in outcome measure effect size ("sparing") as compared to acute exposures. We expected to see minor differences between males and females, with reduced radiosensitivity of females based on the literature. We were surprised on many counts. With regard to inflammatory responses and microglial activation that could lead to synaptic damage, our histological assessment of glial markers indicated that inflammatory responses had largely resolved by 90 days, but we did not look at earlier times when microglial depletion has been shown to be very protective. Limited cytokine analysis from serum at 4 months after proton exposure indicated elevation of two proinflammatory markers but not others. There was evidence for neural cell and synaptic remodeling and membrane-associated damage. In the retina, there was an elevation of the lipid peroxidation marker 4-HNE and reductions in markers associated with the neurovascular niche and blood-retinal barrier involving astrocytes and endothelial cells. There was also some elevation of apoptosis markers, but no significant anatomical evidence of cell death. c-fos expression showed extensive changes in connectivity between brain regions, which is created by synaptic connections that could have been newly formed, deleted, or altered in strength. To determine which gene expression patterns and associated signaling pathways underlay outcomes of the treatment conditions, we used NanoString® technology to profile mRNA expression at 3 months post irradiation of male mice in the orbitofrontal cortex, hippocampus, and cerebellum. A set of 760 genes from the nCounter® Mouse Neuropathology panel for 23 fundamental pathways was examined and characterized according to annotations for: structural integrity, metabolism, neuroinflammation, neuron-glia interaction, plasticity & aging, and neurotransmission. Transcriptomics analysis found that expression patterns varied with radiation dose rate, sex, and brain region. While the magnitude of individual gene differential expression was low, pathway enrichment scores were informative. Principal component analysis of the data showed that brain regions and sexes varied dramatically in expression patterns both without and with radiation exposure. PCA also separated patterns based on exposure to radiation, but with only limited discrimination by dose rate. Transcriptomics data showed gene enrichment changes in pathways associated with neural connectivity in the hippocampus and frontal cortex, and many pathways associated with neurotransmitter synthesis, release, storage, and vesicle trafficking, with opposing trends between sexes and brain regions. The reasons for opposing trends are unclear, but these findings suggest that synaptic remodeling was focused on presynaptic mechanisms. This is the opposite of published electrophysiological findings from the male hippocampus where post-synaptic changes were more salient. Transcriptomics also suggested changes to pathways associated with axon and dendrite structure, and surprisingly, angiogenesis and vascular properties. Another important result from transcriptomics was that the baseline (sham exposure) gene expression patterns for males and females in each brain region were distinct. This argues that each tissue region with its own specific set of cells, signaling pathways, preferred neurotransmitters and excitatory or inhibitory tone responds differently to radiation exposure and its accompanying physical and chemical damage. This argues for the use of behavioral measures which integrate activities of all brain regions, reflecting both residual damage and any stable compensatory responses. DREF estimates were challenging and surprising. We realized that outcome measures were in a variety of native units, such as distance, time, percent, etc., such that intercomparison required transformation to common metrics. Z-scores and Hedges g effect sizes were used to harmonize the data for comparison. Behavioral results immediately disproved the notion that acute exposures would necessarily elicit greater changes than fractionated exposures. Rather, fractionated exposures often lead to greater effects than acute exposures and sometimes acute exposures did not differ significantly from shams. This forced us to rethink the meaning of DREFs in the context of behavior. Whereas traditional radiobiology deals with binary outcomes such as cell death, mutation, and carcinogenesis that can reliably be regarded as deleterious, behavior outcomes are a continuum, and one cannot automatically determine whether the reduction or enhancement of a parameter is good or bad for the organism. Thus, in looking at the ratio of effects between fractionated and acute exposures, it was inappropriate to assign a beneficial or detrimental attribute, but rather just consider the

magnitudes of the ratios. Overall, DREFs for protons and GCRsim were similar for both sexes and ranged from about 1.4 to 1.6. We reduced the number of parameters used for the estimates by applying thresholds for minimum effect size and maximum type I errors (t-test p values). So, our initial hypothesis was incorrect, and both radiation fields led to similar, modest DREF values. These are the first DREF values estimated for behavioral outcomes with charged particle radiation.

Calculation of RBEs used the g values as per DREF. We used the ratio of slopes of GCRsim and gamma responses, assuming linearity in the dose regime explored. We found that RBEs were about 3.9 for all behavioral outcome measures in males and females in response to either acute or fractionated exposures. Thus, the track structure of charged particles leads to cellular and tissue-level changes that are independent of dose rate and sex.

c-Fos+ mapping showed that pre-irradiation of the brain resulted in changes to the way its network components interacted in response to a strong behavioral stimulus. The behavior served to input signals to the network, which propagated signals according to anatomical connections and synaptic states of neurons within the network. Some of these circuits showed activation of c-fos in response to neuron firing patterns within their components. The c-fos connectivity maps clearly illustrated that the functional connectivity patterns were different in terms of reactions to the behavioral stimulus (fear memory recall). Further, the maps showed that the functional connectivity relationships differed as functions of radiation type, dose rate, and sex. Thus, each brain region in each sex responds to the radiation insult in its own unique way, so that overall CNS responses need to be understood collectively using approaches like behavior that integrate over the whole brain. We will now register the c-fos connectivity maps to anatomical connectivity maps to understand whether the c fos connectivity is driven by anatomical relationships or other control systems.

During Covid-19 driven delays in the use of irradiation facilities, we conducted pilot experiments for the chronic mild stress (CMS) model as a surrogate for combined stressors experienced during spaceflight. This well-vetted model delivered mild stress from disruption of cage environment, lighting, social interactions, predator cues, etc. (microstressors) over a 4-week period as a surrogate for the multisensory set of non-radiation space flight stressors. Several anxiety-related outcome measures showed strong increases at 1- and 4-weeks post CMS. Corticosterone levels and two proinflammatory cytokines were also elevated. The successful protocol was designed to be compatible with co-irradiations for a more complete spaceflight simulation, but this was not performed due to scheduling of GCRsim experiments as recommended by NASA. Taken together, the most important conclusion from our study is that responses of the central nervous system to radiation are not continuous quantitative functions of dose rate. Rather, there are complex sets of changes that are both quantitative and qualitative in nature but result in altered behavioral performance. These involve region and sex-specific strategies to restore homeostasis to the affected animal. These strategies may also be radiation quality dependent. The long-term status of the irradiated animal may harbor latent damage and reflect either well-compensated or inappropriately compensated properties that could lead to unwanted consequences under system stress.

Reporting

To date, presentations of experimental findings have been made to the 2019-2025 annual meetings of the Radiation Research Society and International Congress on Radiation Research, the 21st Annual World Congress of Society for Brain Mapping and Therapeutics, the SHINE online NASA training course and the 2020-2025 NASA HRP Investigators' Workshops. Publications and presentations were funded all or in part by the VNSCOR task, and multiple manuscripts are in development that integrate the data.

Bibliography Type:	Description: (Last Updated: 03/26/2026)
Abstracts for Journals and Proceedings	Nelson, G, Saha J. "The space radiation environment and associated human health risks." 71st Radiation Research Society Meeting, San Juan, Puerto Rico, September 21-24, 2025. Abstracts. 71st Radiation Research Society Meeting, San Juan, Puerto Rico, September 21-24, 2025. , Sep-2025
Abstracts for Journals and Proceedings	Dudzinski SO, Harlan CJ, Wang L, Fowlkes NW, Polasek SL, Lin S J-H, Spiotto MT, Nevozhay D, Sony CP, Sokolov KV, Nelson GA, Mao X-W, Bankson JA, Frank SJ. "Effects of simulated space radiation and TP508 as a radiation mitigator on the intraocular immune microenvironment and the systemic immune system." 71st Radiation Research Society Meeting, San Juan, Puerto Rico, September 21-24, 2025. Abstracts. 71st Radiation Research Society Meeting, San Juan, Puerto Rico, September 21-24, 2025. , Sep-2025
Abstracts for Journals and Proceedings	Nelson GA, Jones, T, Stanbouly S. "Dose rate effects of high LET radiation on the mouse nervous system." 71st Radiation Research Society Meeting, San Juan, Puerto Rico, September 21-24, 2025. Abstracts. 71st Radiation Research Society Meeting, San Juan, Puerto Rico, September 21-24, 2025. , Sep-2025
Abstracts for Journals and Proceedings	Nelson G, Jones T, Mao X, Stanbouly S. "Summary of results from space radiation dose rate VNSCOR project." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Beavers M, Blattnig S, Boice JD Jr, Cekanaviciute E, Costes SV, Dauer L, Davis C, Miller DK, Nelson G, O'Banion MK, Tolmachev S. "Radiation-induced Parkinson's disease: evaluating the level of evidence and recommending future work." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Reinsch SS, Nelson GA, Santa Maria SR, Guida P, Sivertz M, Ramanian S, Saha J, Weeks J, Zawaski JA. "NASA's space health impacts for the NASA experience (SHINE) Training program – space radiation curriculum." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025